Statistical Analysis Report

Week 3 Project Deliverable - descriptive statistics, hypothesis testing, correlation, and regression analysis

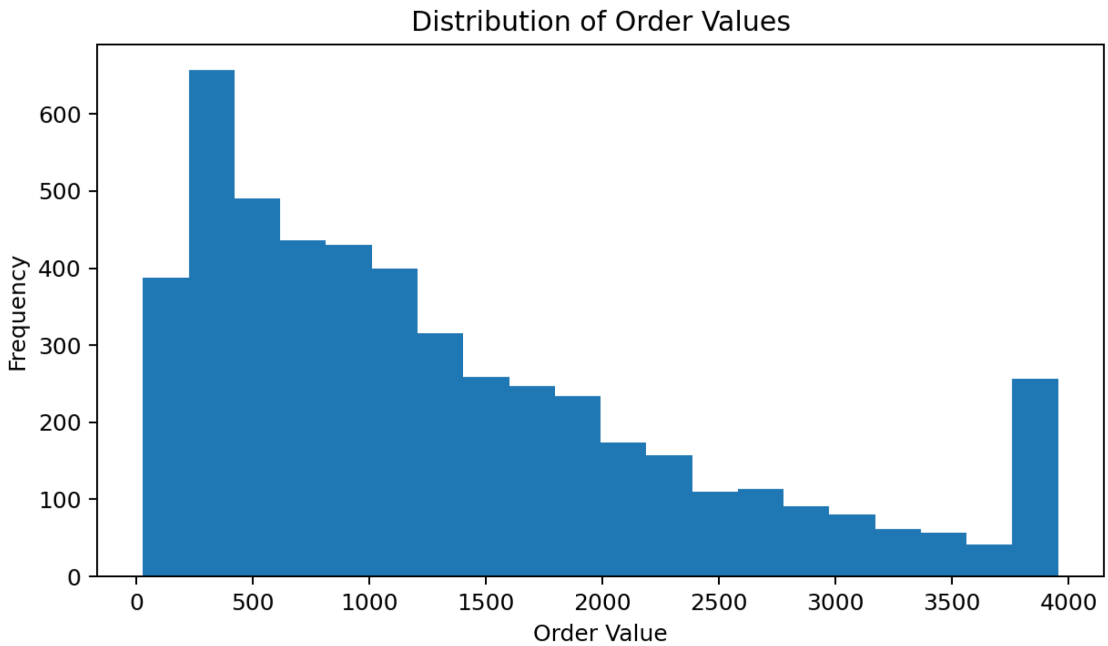
1. Introduction

This report analyzes transaction, customer, product, and A/B testing data to support business decisions with statistical evidence. The analysis covers basic statistics for order values, a simple distribution review, two hypothesis tests, correlation between product price and sales volume, and a linear regression model that predicts monthly revenue using the number of orders.

2. Basic Statistics for Order Values

Metric	Value
Mean	1,338.11
Median	1,061.38
Standard Deviation	1,042.72
Minimum	28.09
Q1	485.49
Q3	1,873.29
Maximum	3,954.98
Skewness	1.005

The mean order value is 1,338.11, while the median is 1,061.38. Because the mean is higher than the median and the skewness is 1.005, the order value distribution is positively skewed. This suggests that most orders are moderate in size, with some larger purchases pulling the average upward.



Distribution analysis shows a right-skewed pattern. The histogram indicates that lower and mid-range order values occur more often than very high order values. This is common in retail-style transaction data, where a smaller number of large purchases increases the spread of the data.

3. Hypothesis Tests

Test 1: Do email customers spend more than non-email customers? (t-test)

Before running the t-test, the customer dataset was checked for two groups: customers with email and customers without email. The dataset contains 10,000 customers with an email address and 0 customers without an email address.

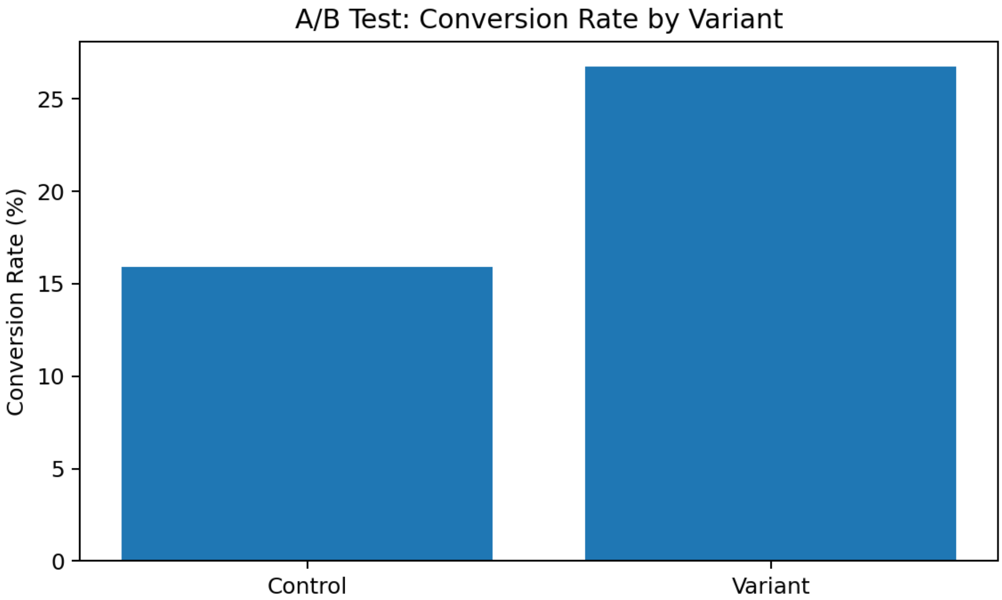
Because all customers in the dataset have an email address, there is no non-email comparison group. Therefore, a valid independent-samples t-test cannot be performed on this dataset. For this test, the p-value is reported as not applicable (N/A), and no statistical conclusion can be made about email vs non-email spending from the available data.

Item	Result
Email customers	10,000
Non-email customers	0
p-value	N/A
Conclusion	Test not applicable because no non-email group exists

Test 2: Does the new checkout have better conversion? (proportion test)

A two-proportion z-test was used to compare conversion rates between the Control checkout and the Variant checkout. This test evaluates whether the observed difference in conversion is large enough to be statistically significant rather than due to random variation.

Group	Users	Converted	Conversion Rate
Control	961	153	15.92%
Variant	1,039	278	26.76%
z-statistic			5.888
p-value			3.903e-09



The Variant checkout conversion rate is 26.76%, compared with 15.92% for the Control. The p-value is 3.903e-09, which is far below 0.05. Therefore, the null hypothesis is rejected and the conclusion is that the new checkout performs significantly better.

4. Correlation Analysis

To understand whether more expensive products tend to sell more or less, product price was compared with product sales volume. Sales volume was calculated as the total quantity sold for each product.

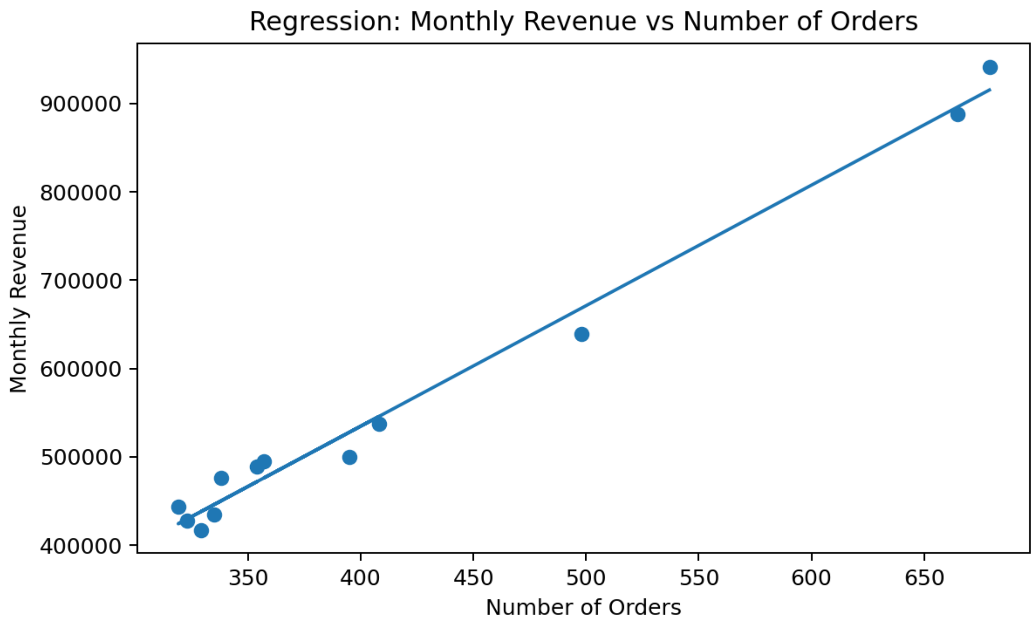
Metric	Value
Correlation between price and sales volume	0.058

The correlation value is 0.058, which indicates a very weak positive relationship between product price and sales volume. In practical terms, price does not appear to be a strong driver of how many units are sold in this dataset. Other factors such as category, brand, promotions, or customer preferences may have a bigger effect.

5. Linear Regression: Monthly Revenue from Number of Orders

A simple linear regression model was used to predict monthly revenue based on the number of orders in each month. The goal is to test whether order volume can explain changes in revenue over time.

Metric	Value
Regression equation	Revenue = -10,660.61 + 1,363.69 x Orders
R-squared	0.986



The coefficient of determination (R-squared) is 0.986. This means that about 98.6% of the variation in monthly revenue is explained by the number of orders. This is a very strong relationship. The fitted slope is positive, showing that as the number of orders increases, monthly revenue also increases.

6. Conclusions and Business Interpretation

The descriptive statistics show that order values are spread out and positively skewed, meaning a small number of high-value orders increase the average. The A/B test result is the strongest statistical finding in the report: the Variant checkout significantly improves conversion relative to the Control checkout.

The email vs non-email spending test could not be completed because the available customer data contains only customers with email addresses. This is a useful data-quality finding because it highlights the importance of having a true comparison group before running a hypothesis test.

The price-to-sales-volume correlation is very weak (0.058), suggesting that product price alone does not strongly determine sales volume. The regression model shows a strong relationship between order count and monthly revenue, with an R-squared of 0.986. This supports the business idea that increasing order volume is likely to increase revenue.

7. Final Summary

- Mean order value = 1,338.11; median = 1,061.38; standard deviation = 1,042.72.

- Order values are positively skewed, so the distribution is not perfectly symmetric.

- Email vs non-email t-test could not be run because all customers in the dataset have an email address.

- Variant checkout conversion (26.76%) is higher than Control (15.92%), with p-value 3.903e-09.

- Correlation between price and sales volume is weak (0.058).

- Linear regression R-squared = 0.986, showing that order count explains most monthly revenue variation.